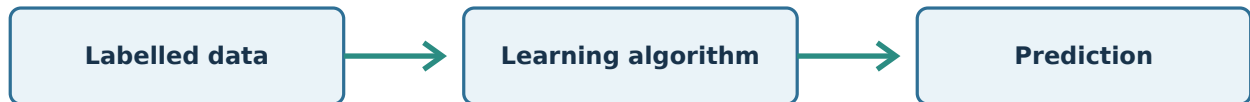


Part A: Conceptual Understanding

Supervised Learning — Questions, Answers and Visuals

1. What are Supervised Learning Algorithms?

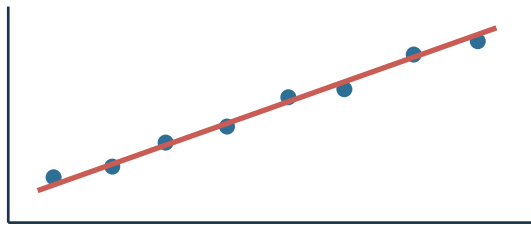
Supervised learning algorithms learn from labelled data, where each input has a known output. They discover patterns and use them to predict outputs for new data. Examples include linear regression, logistic regression, decision trees, and support vector machines.



Supervised learning uses known answers to learn how to predict new ones.

2. Difference between Regression Algorithms and Classification Algorithms.

Regression predicts continuous numerical values, such as house price or temperature. Classification predicts discrete labels, such as spam/not spam or disease/no disease. The key difference is the type of output.



Regression: continuous value



Classification: category

3. Explain Simple Linear Regression.

Simple linear regression models the relationship between one independent variable x and one continuous dependent variable y . It fits the line $y = \beta_0 + \beta_1 x + \epsilon$, where β_0 is the intercept, β_1 is the slope, and ϵ is random error.

Assumptions and Model Fit

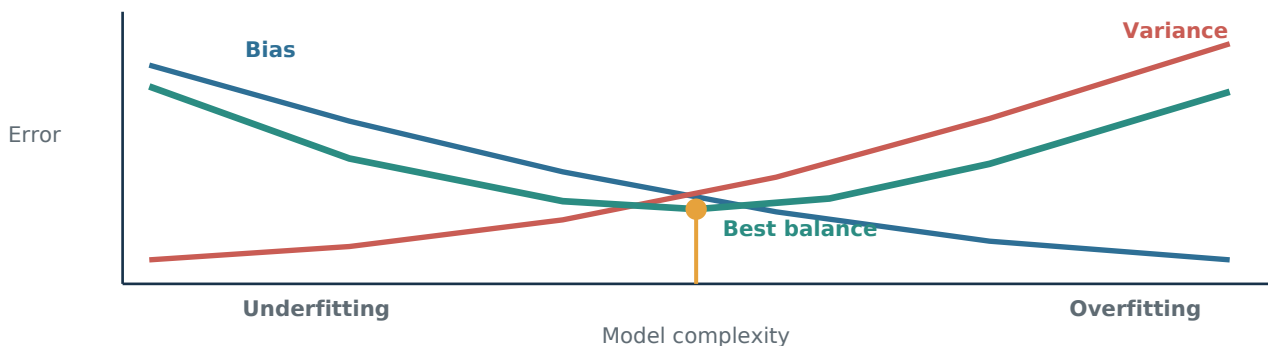
Key ideas for building a model that generalizes well

4. List and explain the assumptions of Linear Regression.

Linearity: predictors have a linear relationship with the outcome. Independence: observations and errors are independent. Homoscedasticity: error variance is constant. Normality: residuals are approximately normal. Predictors should also have no perfect multicollinearity.

5. What is the Bias-Variance Trade-Off?

Bias is error from assumptions that are too simple; variance is error from excessive sensitivity to training data. Greater complexity often reduces bias but increases variance. A good model balances both and performs well on unseen data.



6. Explain Overfitting and Underfitting with examples.

Overfitting means memorizing training noise and failing on new data—for example, a very deep decision tree. Underfitting means being too simple to capture the real pattern—for example, fitting a straight line to strongly curved data.

Underfitting
High bias
Poor on training and test data

Good fit
Balanced complexity
Strong test performance

Overfitting
High variance
Great training, poor test performance